

ORIGINAL INVESTIGATIONS



Biomarker-Based Risk Model to Predict Cardiovascular Mortality in Patients With Stable Coronary Disease

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ABSTRACT

BACKGROUND Currently, there is no generally accepted model to predict outcomes in stable coronary heart disease (CHD).

OBJECTIVES This study evaluated and compared the prognostic value of biomarkers and clinical variables to develop a biomarker-based prediction model in patients with stable CHD.

METHODS In a prospective, randomized trial cohort of 13,164 patients with stable CHD, we analyzed several candidate biomarkers and clinical variables and used multivariable Cox regression to develop a clinical prediction model based on the most important markers. The primary outcome was cardiovascular (CV) death, but model performance was also explored for other key outcomes. It was internally bootstrap validated, and externally validated in 1,547 patients in another study.

RESULTS During a median follow-up of 3.7 years, there were 591 cases of CV death. The 3 most important biomarkers were N-terminal pro-B-type natriuretic peptide (NT-proBNP), high-sensitivity cardiac troponin T (hs-cTnT), and low-density lipoprotein cholesterol, where NT-proBNP and hs-cTnT had greater prognostic value than any other biomarker or clinical variable. The final prediction model included age (A), biomarkers (B) (NT-proBNP, hs-cTnT, and low-density lipoprotein cholesterol), and clinical variables (C) (smoking, diabetes mellitus, and peripheral arterial disease). This “ABC-CHD” model had high discriminatory ability for CV death (c-index 0.81 in derivation cohort, 0.78 in validation cohort), with adequate calibration in both cohorts.

CONCLUSIONS This model provided a robust tool for the prediction of CV death in patients with stable CHD. As it is based on a small number of readily available biomarkers and clinical factors, it can be widely employed to complement clinical assessment and guide management based on CV risk. (The Stabilization of Atherosclerotic Plaque by Initiation of Darapladib Therapy Trial [STABILITY]; [NCT00799903](https://doi.org/10.1016/j.jacc.2017.06.030)) (J Am Coll Cardiol 2017;70:813–26)
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ABBREVIATIONS
AND ACRONYMSGDF-15 = growth
differentiation factor 15hs-cTnT = high-sensitivity
cardiac troponin TLDL-C = low-density
lipoprotein cholesterolNT-proBNP = N-terminal
pro-B-type natriuretic peptide

Coronary heart disease (CHD) remains a major cause of morbidity and mortality worldwide (1,2). Today, patients with stable CHD are recommended standardized secondary prevention measures; such as cessation of smoking; control of blood pressure, lipids, and glucose; revascularization, if myocardial ischemia is present; and platelet inhibition to prevent coronary or stent thrombosis. However,

patients with stable CHD are heterogeneous in their risk of new cardiovascular (CV) events and might benefit from different intensity and duration of preventive treatments. Currently, there are several emerging treatment alternatives, including intense platelet inhibition (3), further cholesterol lowering (e.g., by ezetimibe or proprotein convertase subtilisin/kexin type 9 inhibition) (4,5), and neprilysin inhibition (6), that may need to be applied in selected patients rather than in the broad unselected patient

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Dr. Cannon has received research grants from Arisaph, AstraZeneca, Bristol-Myers Squibb, Boehringer Ingelheim, GlaxoSmithKline, Janssen, Merck, and Takeda; and has received consulting fees from Alnylam, Amgen, Arisaph, Boehringer Ingelheim, Boehringer Ingelheim/Lilly, Bristol-Myers Squibb, GlaxoSmithKline, Kowa, Merck, Takeda, Lipimedix, Pfizer, Regeneron, and Sanofi. Dr. Granger has received grants and consultancy fees from AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, GlaxoSmithKline, Pfizer, Sanofi, Takeda, The Medicines Company, Daiichi-Sankyo, Janssen, and Bayer; has received grants from Medtronic Foundation and Armetheon; has received consultancy fees from Abbvie, Boston Scientific, Eli Lilly, Hoffman-La Roche, Salix Pharmaceuticals, Gilead, Medtronic, Novartis, Sirtex, and Versone; and has received a research grant from Novartis. 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Dr. Siegbahn has received institutional research grants from AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb/Pfizer, and GlaxoSmithKline. Dr. März has received grants and personal fees from Siemens Diagnostics, Aegerion Pharmaceuticals, Amgen, AstraZeneca, Danone Research, Sanofi/Genzyme, Sanofi, BASF, and Numares AG; has received personal fees from Alexion, Hoffmann LaRoche Pharma, and MSD; has received grants from Roche Diagnostics and Abbott Diagnostics; and is an employee of Synlab Holding Deutschland. Dr. Wallentin has received institutional research grants, consultancy fees, lecture fees, and travel support from Bristol-Myers Squibb/Pfizer, AstraZeneca, GlaxoSmithKline, and Boehringer Ingelheim; has received institutional research grants from Merck & Co. and Roche; has received consultancy fees from Abbott; and holds 2 patents involving GDF-15. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose. Iftikhar J. Kullo, MD, served as Guest Editor for this paper.

population. To appropriately tailor the intensity of secondary preventive treatments, there is a need for improved risk stratification tools for patients with stable CHD.

At present, there is no generally accepted or used model for risk stratification in patients with stable CHD. CALIBER (Cardiovascular Disease Research Using Linked Bespoke Studies and Electronic Health Records) is a previously presented model for stable CHD, which included 22 variables and yielded good discrimination for all-cause mortality (7). However, the large number of variables would potentially preclude implementation in everyday clinical practice. Another model was reported from the LIPID (Long-Term Intervention with Pravastatin in Ischemic Disease) trial, where a combination of 10 clinical variables, including total and low-density lipoprotein cholesterol (LDL-C), proved useful in identifying patients with more benefit from statin treatment (8).

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During recent years, several circulating biomarkers have been found to carry prognostic information in stable CHD. Higher levels of cardiac biomarkers, such as N-terminal pro-B-type natriuretic peptide (NT-proBNP) and cardiac troponins, measured with high-sensitivity assays, are associated with a higher risk of CV events in patients with stable CHD (9–14). Several other biomarkers are prognostic in this setting: markers of inflammatory activity, such as high-sensitivity C-reactive protein (hsCRP) (15), interleukin (IL)-6 (16), renal dysfunction (cystatin-C) (17), and cellular stress (growth differentiation factor [GDF]-15) (17), as well as lipoprotein-associated phospholipase A₂ (Lp-PLA₂) (18).

The objectives of this study were to evaluate the incremental prognostic value of high-sensitivity cardiac troponin T (hs-cTnT) and NT-proBNP, in addition to clinical information; then, in comparison with other biomarkers, to develop and validate a clinically useful biomarker-based prediction model for CV death and other CV events in patients with stable CHD.

METHODS

The biomarker substudy of the STABILITY (Stabilization of Atherosclerotic Plaque by Initiation of Darapladib Therapy) trial was used to identify the most important biomarkers and derive the prediction model. Briefly, the STABILITY trial compared darapladib, a selective inhibitor of Lp-PLA₂, with placebo in 15,828 patients with stable CHD on optimal secondary preventive treatment, regarding CV events. Patients were enrolled at 663 centers in 39 countries

between December 2008 and April 2010. Patients were eligible if they had experienced at least 1 of the following: myocardial infarction (MI), percutaneous coronary intervention, or coronary artery bypass graft surgery; or demonstrated multivessel coronary artery disease (CAD) at a coronary angiogram (Online Table 1). Additionally, patients had to be treated with a statin and have at least 1 of the following risk enrichment criteria: age >59 years, diabetes mellitus (DM), high-density lipoprotein cholesterol (HDL-C) <40 mg/dl (1.03 mmol/l), smoking within the last 3 months, renal dysfunction, or concomitant cerebrovascular or peripheral arterial disease (PAD) (Online Table 2). Patients were excluded if they had experienced an MI during the last month, if they had coronary revascularization during the last 3 months, or if there was a pre-planned coronary revascularization procedure. Other exclusion criteria included severe heart failure (HF) (New York Heart Association functional class III or IV), severe renal failure, and other noncardiac comorbidities listed in Online Table 1. Follow-up was performed by regular outpatient visits, with a median follow-up of 3.7 years (interquartile range: 3.5 to 3.8 years) until 2014. Additional details regarding study design and principal results have been published previously (19,20). The trial was approved by the relevant institutional review boards and performed in accordance with the Declaration of Helsinki. All patients provided written informed consent to participate.

The LURIC (Ludwigshafen Risk and Cardiovascular Health) prospective observational study was used for validation. The LURIC study included 3,279 patients scheduled for coronary angiography for nonacute chest pain between July 1997 and January 2000. Patients presenting with unstable angina, non-ST-segment elevation MI, or ST-segment elevation MI were excluded from the current evaluation. Patients with severe diseases other than stable CAD were also excluded. Thus, the validation cohort consisted of 1,547 individuals with stable CAD and available analyses of biomarkers at baseline. All subjects were followed until their death or at least for 12 years, including assessment of vital status and medical history by questionnaire, and cause of death by death certificate. The study was performed in accordance with the Declaration of Helsinki and was approved by the local ethics committee. All participants gave their written informed consent prior to inclusion (21).

BIOMARKER ANALYSES. In patients included in the STABILITY biomarker substudy, blood samples were obtained at randomization. For both STABILITY and

TABLE 1 Baseline Characteristics

	STABILITY (N = 13,164)	LURIC (N = 1,547)
Age, yrs	65.0 (59.0-71.0)	64.7 (58.0-71.1)
Male	10,756 (81.7)	1,170 (75.6)
Body mass index, kg/m ²	28.4 (25.6-31.8)	—
Current smoker	2,395 (18.2)	247 (16.0)
Hypertension	9,418 (71.5)	1,197 (77.4)
Diabetes	5,141 (39.1)	542 (35.0)
Previous MI	7,748 (58.9)	707 (45.7)
Previous revascularization	9,860 (74.9)	574 (37.3)
Multivessel CHD	1,819 (13.8)	—
Polyvascular disease	1,998 (15.2)	—
Peripheral arterial disease	1,128 (8.6)	191 (12.3)
hsCRP, mg/l	1.3 (0.6-3.1)	2.8 (1.2-6.7)
hs-cTnT, ng/l	9.3 (6.2-14.2)	10.0 (6.0-19.0)
GDF-15, ng/l	1,256 (920-1,829)	—
NT-proBNP, ng/l	172 (83-373)	299 (113-875)
Cystatin C, mg/l	1.0 (0.9-1.2)	0.9 (0.8-1.1)
LDL cholesterol, mmol/l	2.1 (1.6-2.6)	3.0 (2.4-3.6)
HDL cholesterol, mmol/l	1.2 (1.0-1.4)	1.0 (0.8-1.1)
Triglycerides, mmol/l	1.5 (1.1-2.1)	1.7 (1.2-2.3)
Hemoglobin, g/l	144 (134-152)	140 (130-150)
White blood cells, x 10 ⁹ /l	6.6 (5.5-7.8)	6.7 (5.6-8.0)
Glomerular filtration rate, ml/min/1.73 m ²	74 (62-87)	87 (73-97)

Values are median (interquartile range) or n (%).

CHD = coronary heart disease; GDF-15 = growth differentiation factor 15; HDL = high-density lipoprotein; hsCRP = high-sensitivity C-reactive protein; hs-cTnT = high-sensitivity cardiac troponin T; LDL = low-density lipoprotein; LURIC = Ludwigshafen Risk and Cardiovascular Health; MI = myocardial infarction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; STABILITY = Stabilization of Atherosclerotic Plaque by Initiation of Darapladib Therapy.

LURIC, plasma samples were stored frozen until biomarker analyses were performed.

In the STABILITY cohort, plasma levels of hs-cTnT and NT-proBNP were determined using electrochemiluminescence immunoassays. GDF-15 was measured with a GDF-15 pre-commercial assay (22). These biomarker assays were performed at the Uppsala Clinical Research Center Laboratory at Uppsala University, Uppsala, Sweden. Lp-PLA₂ activity was measured by the manufacturer in an automated enzyme assay system. For hsCRP, analysis was done using a 2-site particle-enhanced immunonephelometry sandwich assay. The hsCRP analyses, as well as routine biochemical analyses (including lipids, hematology, and creatinine) were performed at a central laboratory. Estimated glomerular filtration rate was determined based on creatinine levels using the Chronic Kidney Disease Epidemiology Collaboration formula. The biomarker methodology and results from the LURIC study have been published previously (23).

OUTCOMES. In the STABILITY cohort, definitions of all outcome events were pre-specified, as previously described (19), and events were adjudicated by an

independent clinical events committee. We evaluated the model's and the included biomarkers' prognostic performance for the primary outcome of CV death, as well as across the following clinically important secondary outcomes: major adverse cardiovascular and cerebrovascular events (MACCE), defined as the composite of CV death, stroke, and MI; major coronary events (the composite of coronary death, MI, and urgent revascularization); all-cause death; MI; stroke; HF; the composite of CV death and HF; and the composite of MACCE and HF.

STATISTICAL ANALYSIS. Demographics and other baseline characteristics were compared across hs-cTnT and NT-proBNP quartile groups, using Kruskal-Wallis tests for continuous variables and chi-square tests for categorical variables. Natural logarithmic (log) transformations were performed for continuous variables with skewed distributions when appropriate. Unadjusted associations between the continuous biomarker levels and all outcomes were assessed and presented as restricted cubic spline plots for the biomarkers included in the final model (i.e., hs-cTnT and NT-proBNP). The relation between these biomarker levels and outcomes were also presented as Kaplan-Meier curves. We performed adjusted analyses of the associations between these biomarkers and all outcomes using Cox proportional hazards models with each biomarker categorized in quartile groups, adjusting for age, systolic blood pressure, and body mass index (BMI) entered as restricted cubic splines, and randomized for treatment, sex, hypertension, geographic region, prior MI, prior revascularization (by percutaneous coronary intervention or coronary artery bypass graft surgery), prior multivessel CHD, DM, smoking, polyvascular disease, renal dysfunction, and all other biomarkers. The adjusted hazard ratios with 95% confidence intervals are presented as forest plots.

Model development and validation. For development of a clinical prediction model, the candidate variables were: age, sex, smoking, BMI, hypertension, previous MI, previous revascularization, previous multivessel CHD, DM, previous stroke, HF, PAD (defined as current intermittent claudication with objective evidence of vascular origin; history of stenting or surgery for PAD, including amputation due to vascular disease; or ankle-brachial index <0.9 in at least 1 ankle), anemia, and the following biomarkers: hs-cTnT, hsCRP, NT-proBNP, LDL-C, HDL-C, GDF-15, Lp-PLA₂, IL-6, triglycerides, white blood cells, and estimated glomerular filtration rate. Of patients with available biomarkers, only 27 patients (0.2%) had missing values regarding clinical variables

in the derivation cohort. Hence, we proceeded with a complete-case analysis without any attempts of imputation.

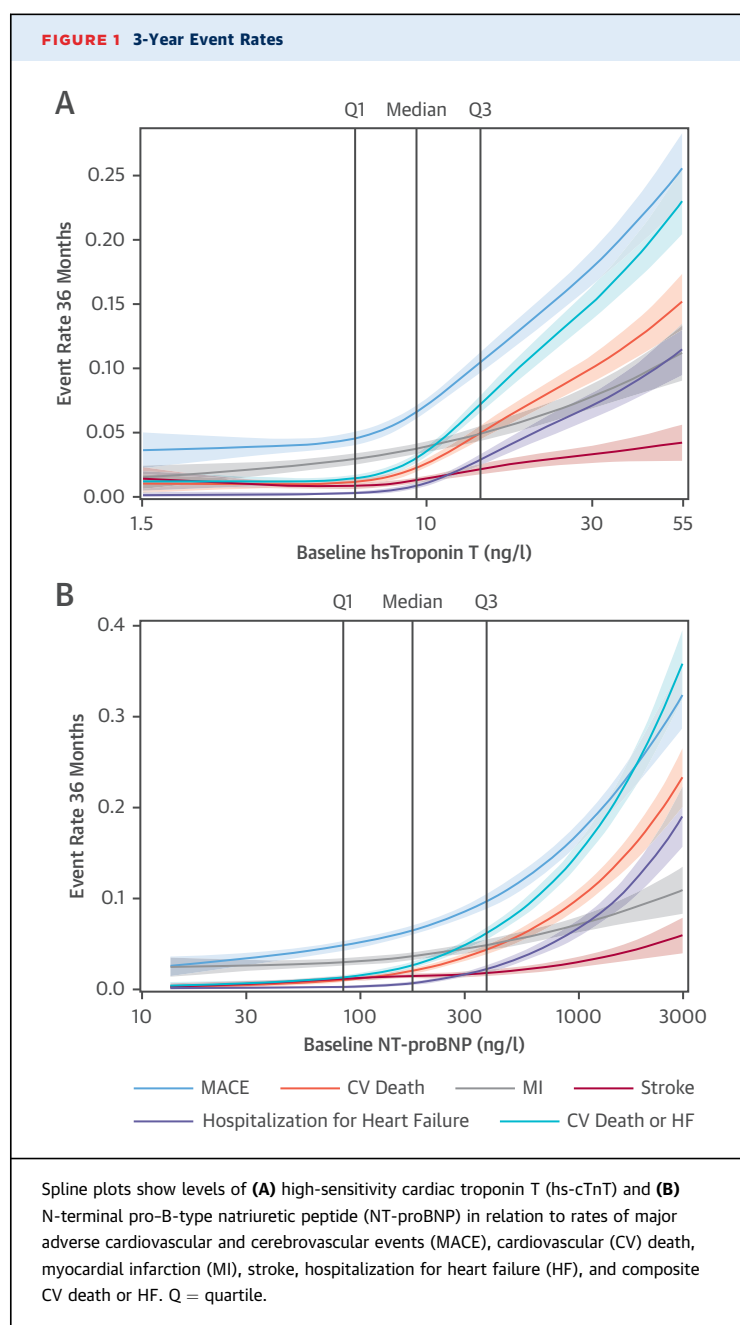
We initially fitted a rich Cox regression model that included all candidate variables. Biomarker levels below the limit of detection were set to one-half of that level. Based on inspection of the distribution of continuous variables, the following variables were log-transformed: hs-cTnT, NT-proBNP, LDL-C, HDL-C, GDF-15, triglycerides, white blood cells, hsCRP, and IL-6. All continuous variables were included as restricted cubic splines, with 4 knots at the 5th, 35th, 65th, and 95th percentiles, to allow for nonlinear associations. The global test of any nonlinear association was significant ($p = 0.004$), so all nonlinear terms were retained in the full model.

For a more clinically useful prediction model, we aimed to approximate the full model by a smaller model. Each predictor's contribution in the full model was measured as the partial chi-square statistic minus the predictor degrees of freedom. Variables for the final smaller model were selected based on their importance in the full model and, among the highly ranked (top 10) markers, by prioritizing previously well-established markers of high risk. These variables were then fitted to the predictions (on the linear predictor scale) from the full model instead of to the outcome. In this manner, the full model was approximated by the subset of the variables and the risk of overfitting was reduced. The final model is represented by a nomogram.

Clinical usefulness was evaluated using decision curve analysis, by estimating the net benefit of using the model to risk stratify patients according to different decision thresholds of 1-year risk of CV death, compared with the 2 alternatives of assuming that none or all will be at high risk, as well as to using a model based only on clinical variables.

Internal validation of the final model was done using 100 bootstrap samples. External validation was performed by applying the final model to data from 1,547 patients included in the LURIC study. Discrimination was assessed with Harrell's c-index for the endpoint used in the development of the model (CV death), as well as across other important endpoints.

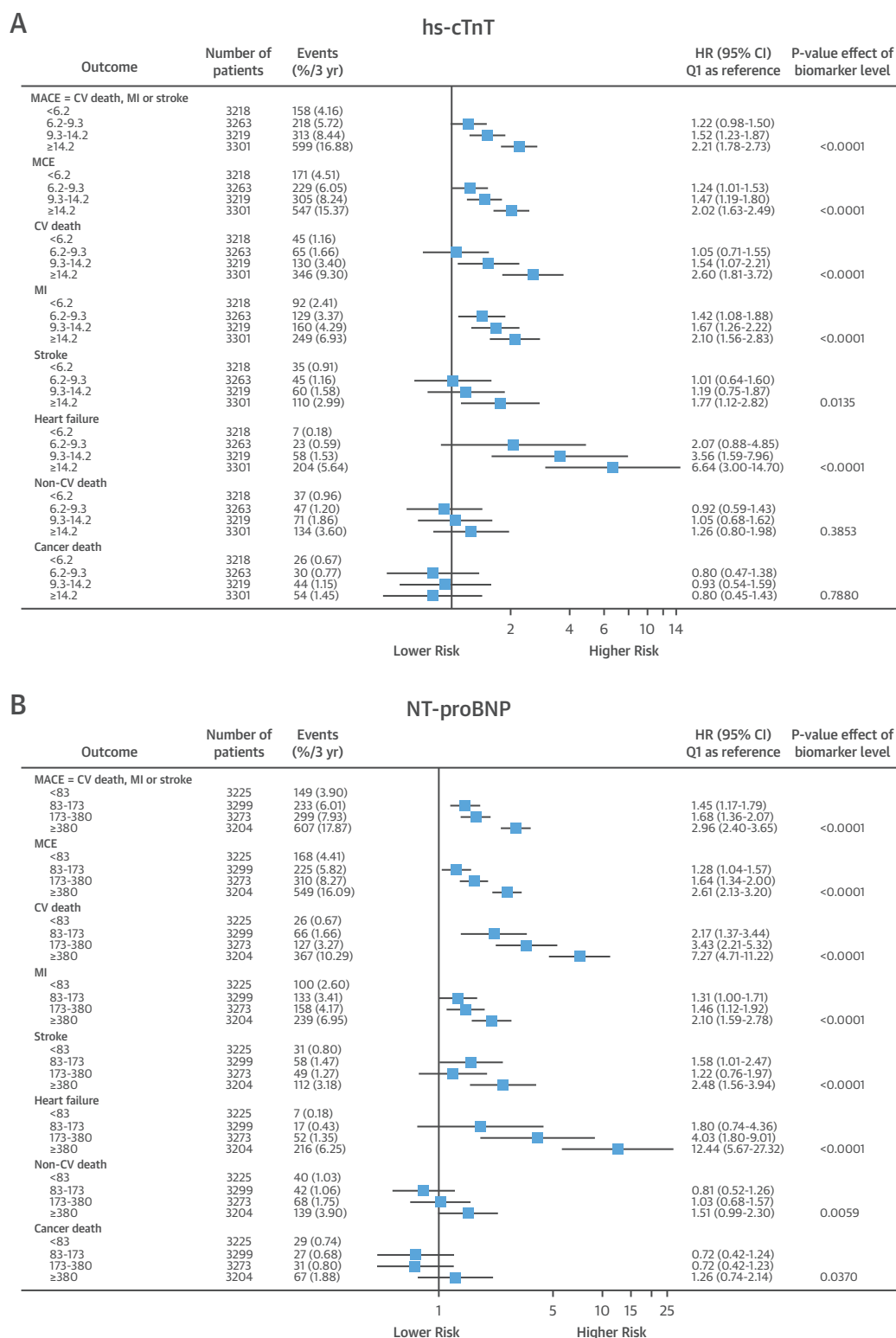
Calibration (i.e., the agreement between observed outcomes and predictions) was assessed graphically, with calibration plots. The development and reporting of this prediction model followed the TRIPOD (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis) statement (24); the TRIPOD checklist can be found in the [Online Appendix](#). SAS version 9.4 (SAS Institute Inc.,



Cary, North Carolina) and R version 3.2.4 (R Foundation for Statistical Computing, Vienna, Austria) were used for all analyses, using the rms package (25) for R. A web calculator was developed using the shiny package (26).

RESULTS

In the derivation cohort, there were 13,164 patients who fulfilled the inclusion criteria of this study ([Online Figure 1](#)). The clinical characteristics are

FIGURE 2 Risk of Events

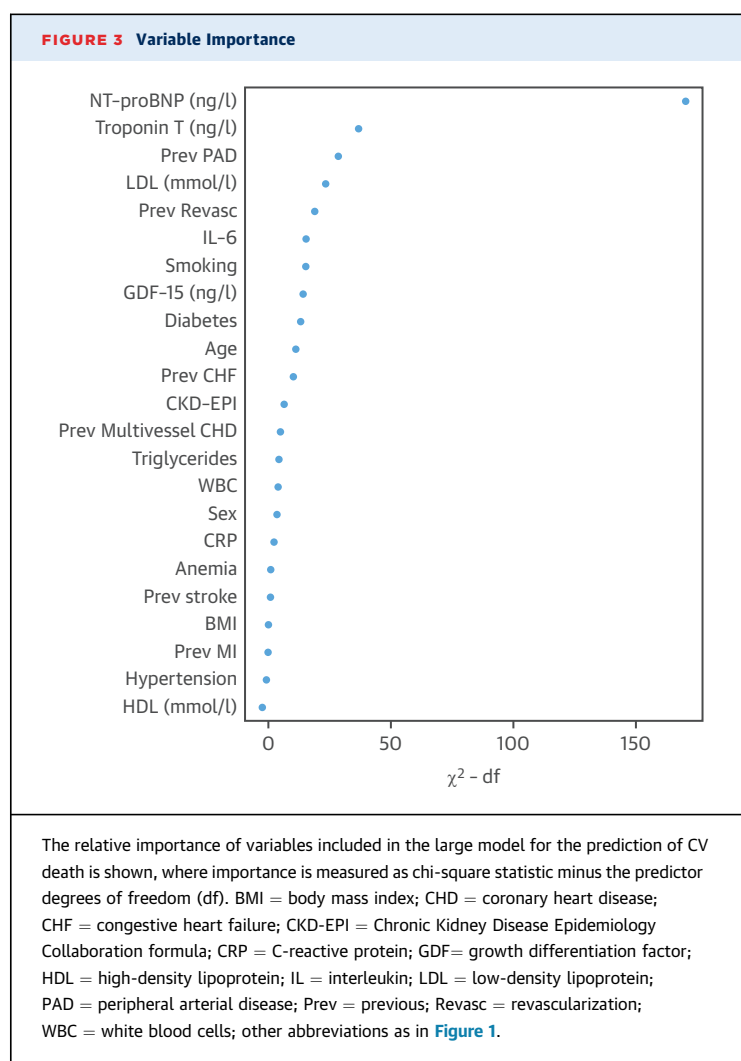
The effect of quartile groups of (A) hs-cTnT and (B) NT-proBNP levels at baseline on outcome was adjusted for all clinical characteristics, risk factors, and biomarkers. CI = confidence interval; HR = hazard ratio; MCE = major coronary event; other abbreviations as in Figure 1.

shown in [Table 1](#) (with extended baseline characteristics in relation to quartiles of NT-proBNP and hs-cTnT shown in [Online Table 3](#)). Several clinical risk markers, such as age, BMI, hypertension and DM, renal dysfunction, prior revascularization, extent of vascular disease, hemoglobin, and geographical region, showed independent associations with NT-proBNP and hs-cTnT levels in multivariable analyses ([Online Table 4](#)). Also, there were significant correlations between NT-proBNP and hs-cTnT as well as other measured biomarkers ([Online Table 5](#)), with the strongest correlation between hs-cTnT and NT-proBNP (Pearson $r = 0.48$).

During the whole course of the study (median follow-up: 3.7 years), there were 591 CV deaths (1.27%/year), 640 MIs (1.41%/year), and 255 strokes (0.55%/year), amounting to 1,305 cases (2.89%/year) for any of these events. There were 296 cases (0.64%/year) of hospitalization for HF, and 291 (0.63%/year) of non-CV death including 154 cases (0.33%/year) of cancer death during follow-up ([Online Table 6](#)). Increasing levels of both hs-cTnT and NT-proBNP were strongly and gradually associated with increasing rates of all major CV adverse events, including CV death, nonfatal MI, nonfatal stroke, hospitalization for HF, and the composites of these events in the unadjusted analyses ([Figure 1](#), [Online Figure 2](#)). After adjustment for the other biomarkers in addition to clinical variables, increasing quartile groups of hs-cTnT and NT-proBNP were associated with increasing risk of all CV events ([Figure 2](#)). There were no statistically significant associations between hs-cTnT or NT-proBNP and non-CV or cancer death ([Figure 2](#)).

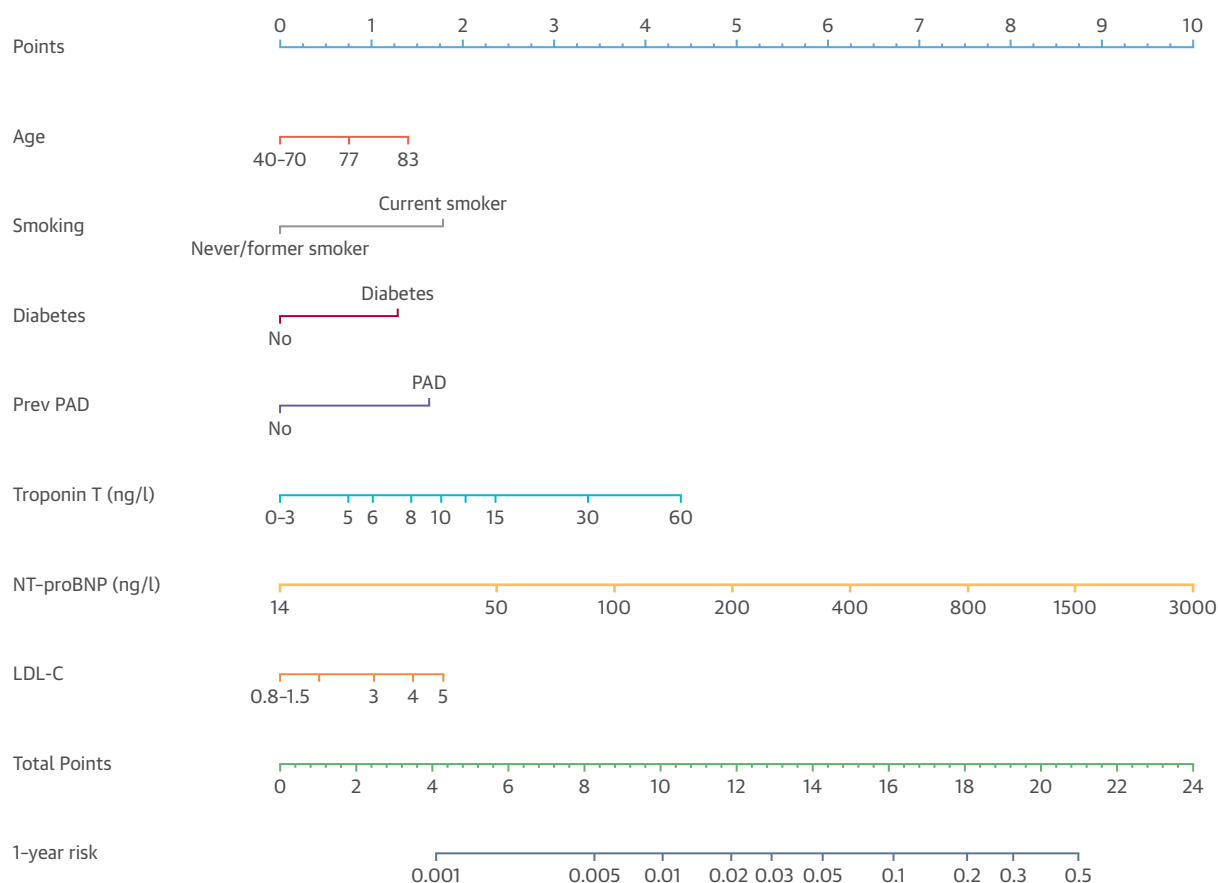
The levels of NT-proBNP and hs-cTnT consistently had larger prognostic value than any clinical variable and all other biomarkers concerning all CV outcomes, as compared using the partial chi-square statistic minus the predictor degrees of freedom. Other variables that significantly contributed to discrimination concerning most CV outcomes were age; DM; smoking; prior PAD; and the biomarkers LDL-C, GDF-15, and IL-6.

Including all candidate variables, the c-index for the full model for prediction of CV death was 0.83. The bootstrap-validated, optimism-corrected c-index was 0.82, and the optimism-corrected calibration slope was 0.94, both indicating only modest overfitting. Each variable's importance in the full model is illustrated in [Figure 3](#). The full model was approximated, among the 10 prognostically most important variables, by selecting the 3 most important biomarker variables (NT-proBNP, hs-cTnT, and LDL-C) and thereafter, the 4 best-established clinical risk factors



(smoking, DM, PAD, and age). The addition of any or all remaining 3 of the top 10 variables only marginally improved the discriminatory value, which was considered too limited to warrant inclusion in the final model (data not shown). As a further approximation, based on visual inspection, the association between each of the included continuous variables and the log relative hazard of CV death was assumed to be linear. The final model approximated 86% of the full model and is represented as a nomogram in [Figure 4](#). With this small final model, the c-index was 0.81. When using this model, derived for CV death, to predict other important outcomes, the discriminatory ability was similarly good; for example, regarding HF (c-index: 0.87), CV death and HF (c-index: 0.81), and MACCE (c-index: 0.71) ([Online Table 7](#)). There was no difference in treatment effect of the randomized treatment in relation to predicted risk (data not shown). In a sensitivity analysis excluding patients

FIGURE 4 Nomogram for the Final Model



Each variable corresponds to a **point (top)**. These points are then summed to translate into a 1-year risk of CV death (**bottom**). LDL-C = low-density lipoprotein cholesterol; other abbreviations as in [Figures 1 and 3](#).

with known history of HF, the prognostic model and its predictive performance were almost identical to those of the full population.

The clinical characteristics of the validation and derivation cohorts were well comparable ([Table 1](#)). Compared with the LURIC cohort, the STABILITY cohort had somewhat higher rates of previous MI and previous revascularization, but lower rates of PAD. The hs-cTnT levels were similar, but the NT-proBNP and LDL-C levels were somewhat lower in the STABILITY cohort. Applying the small model derived from the STABILITY cohort, the c-index for CV death was 0.78 in the validation cohort. The small model also showed good calibration, where predicted and actual event rates corresponded well ([Figure 5](#)) in both the development and validation cohorts. The continued unchanged event rates for up to 12 years in the LURIC cohort indicated that the

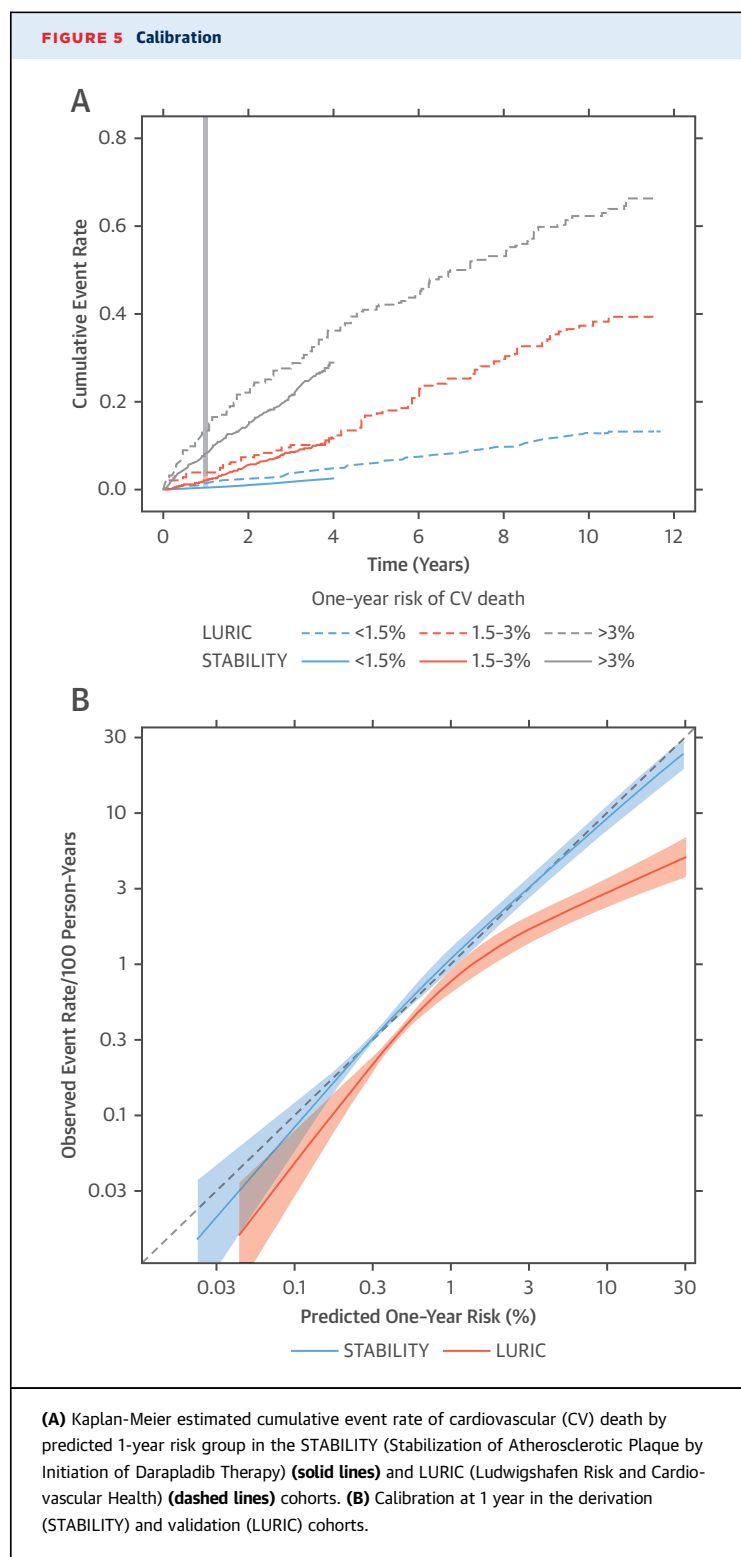
risk stratification by the small model was consistent far beyond the 5-year maximum follow-up in the derivation cohort.

To assess the usefulness of the model, we performed a decision curve analysis. The results demonstrated that for decision thresholds higher than 0.5%/year, the final biomarker-based model was substantially better than a model including only clinical variables. For example, for a threshold of a 5% risk of CV death, compared with a model comprising clinical variables alone, using the biomarker-based model would identify 2 additional cases without identifying any additional false positives in a population with 12.7 CV deaths/1,000 person-years ([Figure 6](#), [Online Table 8](#)). An example on how the clinical prediction model could be implemented in a web-based application is shown in [Online Figure 3](#).

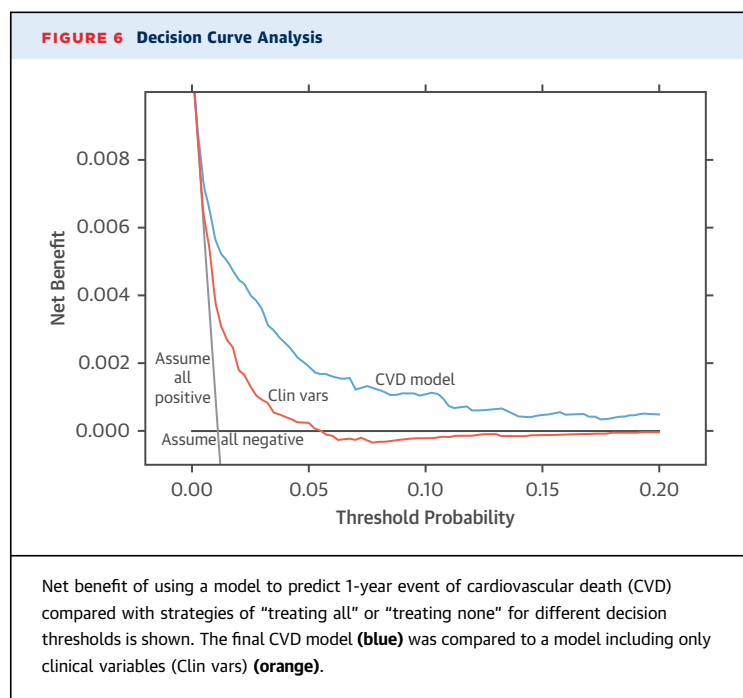
DISCUSSION

In this study, we showed that the cardiac biomarkers NT-proBNP and hs-cTnT had greater prognostic value than any clinical variable and any other biomarker for all CV outcomes, and that they provided incremental predictive information, even in the presence of all clinical variables and information from multiple other prognostic biomarkers. Although other biomarkers provided independent prognostic information, their incremental value for discrimination of risk was limited in the presence of NT-proBNP, hs-cTnT, and LDL-C, plus a small subset of clinical variables. Based on these findings, we derived and internally and externally validated a novel biomarker-based model for the prediction of CV death in patients with stable CHD containing age (A), the biomarkers (B) NT-proBNP, hs-cTnT, and LDL-C, and the clinical variables (C) smoking, DM, and PAD (**Central Illustration**). Despite the small number of variables in this novel ABC-CHD model, it provided robust prediction of CV death and seemed useful for other CV outcomes, including hospitalization for HF. The ABC-CHD model was well calibrated in an external observational cohort of patients with stable CHD. Based on the outcomes in the validation cohort, it seemed as the score might predict the risk of events far beyond the 5-year period of follow-up of the derivation cohort. The model could be easily implemented into clinical practice with a web-based application ([Online Appendix](#)) that could be accessed from computers or mobile devices without any need for integration with electronic health records (EHRs).

So far, there are no generally accepted and used prognostic tools for evaluation of outcomes in patients with stable CHD. Previously, the VILCAD (Vienna and Ludwigshafen CAD) risk score for stable CHD, based mainly on clinical variables, was proposed (27). The VILCAD score was recently supplemented with the addition of biomarkers, which improved prediction in the same cohort (28). The VILCAD development cohort involved a comparatively small sample of patients recruited at a single center between November 1999 and August 2000, which was validated in the same cohort (LURIC) as the current study (recruited between July 1997 and January 2000). A prediction model from the REACH (Reduction of Atherothrombosis for Continued Health) registry (with patients enrolled between December 2003 and June 2004) has been developed as well, including 12 clinical variables. This model showed acceptable discrimination, but was not validated externally (29). Another model developed in



912 patients with stable CHD from the Heart and Soul study (recruited between 2000 and 2002), and validated in 2,876 patients from the PEACE (Prevention of Events with Angiotensin-Converting Enzyme



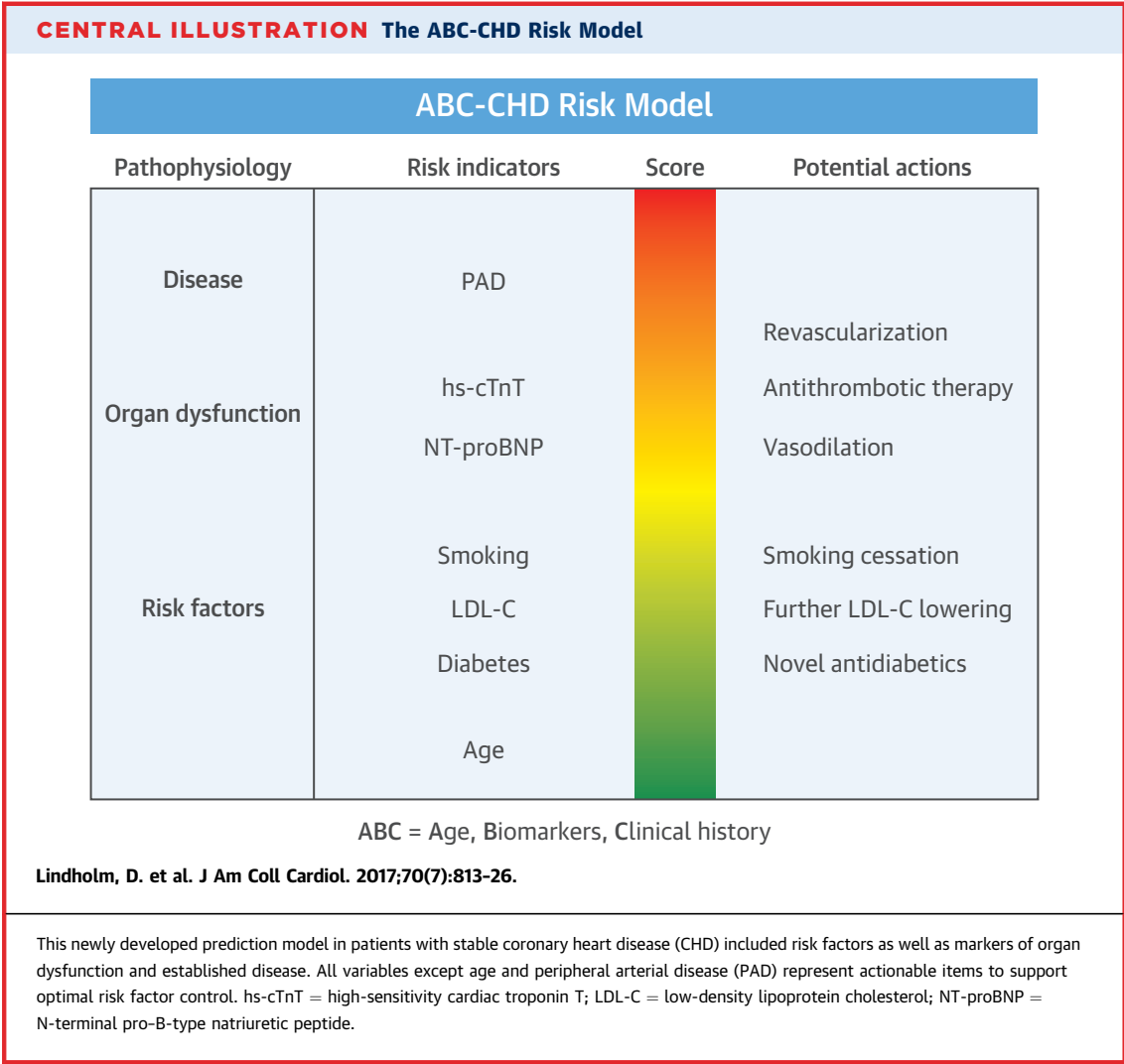
Inhibition) trial (which randomized patients with stable CHD and left ventricular [LV] ejection fraction >40% to trandolapril or placebo between 1996 and 2000), also included hs-cTnT and NT-proBNP as key predictors, in addition to urinary albumin:creatinine ratio and current smoking (30). Comparing the reported results of that model with the ABC-CHD score concerning the discrimination of MACCE indicated similar c-indexes in the respective derivation cohorts. This similarity is likely attributable to the dominating importance of NT-proBNP and hs-cTnT, which were common to both models.

However, using the Heart and Soul model, both discrimination and calibration deteriorated substantially in the validation cohort, suggesting that this model was overfitted to the derivation dataset. In contrast, the ABC-CHD model showed consistent discrimination in both the derivation and validation cohorts. The better performance of the ABC-CHD model might be explained by choice of endpoint (CV death), more included variables, and possibly other factors, such as differences in baseline risk. For atherothrombotic events, such as MI and stroke, our model's discriminatory ability was more modest (c-index of 0.65 for both, although the model was not optimized for prediction of these events), which could be attributable to the stronger association between the biomarkers and CV death and HF compared with between the biomarkers and atherothrombotic events. Recently, the CALIBER model

for prognostication of patients with stable CHD was developed and validated in 2 large cohorts of patients in the National Health Service in the United Kingdom. The CALIBER model was based on 22 variables from patient health records and provided good discrimination and calibration concerning outcomes in patients with stable CHD (7). The CALIBER score will, however, be difficult to implement in everyday practice, unless integrated into the EHR system. However, if integration of decision support tools and EHRs is achieved, the preferred method might be to implement more advanced machine-learning algorithms (31). The ABC-CHD model, with its small number of variables, provides a tool that might be easier to use in a busy clinical setting.

In contrast to the previous risk scores, the derivation cohort for the present ABC-CHD model was based on a contemporary large multinational multicenter population of patients with stable CHD representing a large diversity of patients and health care around the globe. Our methodology focused on identifying a minimal number of variables for risk assessment based on comparing previously documented biomarkers and clinical variables with prognostic value in this setting. The study thereby succeeded in showing that the 2 generally available cardiac biomarkers, NT-proBNP and hs-cTnT, provided stronger prognostic information than any clinical variable or other biomarker. These findings were concordant with previous work from the LURIC cohort, in which the combination of hs-cTnT and NT-proBNP was also superior to the use of either biomarker alone for risk stratification of CV death (23). By including these 2 biomarkers in the score, it was possible to retain 86% of the prognostic value of all clinical and biomarker variables by using a subset of only 7 variables for the score. In addition to the excellent internal validation and calibration, the ABC-CHD score showed similar discrimination and calibration in the external LURIC cohort. These findings supported the use of the ABC-CHD score, which should be generalizable to other cohorts.

The most important prognostic variables in our model were the 2 cardiac biomarkers, where the (by far) strongest predictor for CV death was NT-proBNP. NT-proBNP, as well as BNP, is released into the circulation in response to cardiomyocyte stretch (32) or induced by vasoactive substances, such as angiotensin-II and endothelin-I (33,34). NT-proBNP and BNP are considered cardioprotective by acting on the kidneys to inhibit sodium reabsorption (with subsequent natriuresis and diuresis), on vascular smooth muscle cells (inducing relaxation, thereby



reducing preload), and on the renin-angiotensin-aldosterone system (35). Recent evidence implied a role of NT-proBNP also in metabolic pathways, including lipolysis and regulation of blood glucose levels, which are important in the pathophysiology of CHD (36). Cellular expression and increased levels of NT-proBNP were strongly associated with major cardiac events and death in stable CHD, as demonstrated in this and several prior studies (9-11,13). As an indicator of myocardial dysfunction, NT-proBNP might be a modifiable risk factor, for example, by pre-load and after-load-reducing agents, such as diuretics, angiotensin-converting enzyme or angiotensin-II inhibitors, or even novel agents such as sacubitril/valsartan (6). Given the substantial contribution of NT-proBNP to our model, it could be hypothesized that part of the signal constituted detection of patients with asymptomatic LV dysfunction.

Unfortunately, imaging of LV function was not available to illuminate that.

Using a high-sensitivity assay, troponin T, a protein of the contractile units of cardiomyocytes with nearly 100% cardiac specificity (37), was the second most important marker of risk in this study. As demonstrated previously, elevated troponin T levels are associated with increased risk of cardiac mortality and morbidity in patients with stable CHD as well as the general population (12,14,22,23,38,39). Strikingly, in the PEACE trial, hs-cTnT levels were strongly associated with CV death and HF, but not with MI (12); the reasons for this result have been debated, but not elucidated. However, several mechanisms other than myocardial ischemia have been proposed to cause troponin elevations, such as release of troponin degradation products into the circulation (40), release of troponins from viable cardiomyocytes

in response to stretch (41), microvascular disease (42), and other pathophysiological processes such as apoptosis (43). Recent imaging evidence, however, suggested an association between high-sensitivity troponin levels and presence of high-risk coronary plaques in patients with stable CHD (44), which should be further studied. Slight elevation of troponin was found in a proportion of patients with chronic HF, in whom detectable hs-cTnT predicted adverse outcomes (45), even below the commonly used diagnostic cutoff. In the present study, for instance, where biomarkers were considered continuous predictors as opposed to using specific cut-offs, the risk of CV mortality and morbidity started to increase at hs-cTnT levels well below the commonly used cutoff of 14 ng/l (already at about 10 ng/l). A similar pattern could be discerned for NT-proBNP, with a continuous association (especially for CV death and HF) at low levels (about 100 ng/l upward).

In addition to more established and routinely available biomarkers included in this study (LDL-C, hs-cTnT, and NT-proBNP), there are emerging biomarkers such as ceramides (46) or multiple other proteins (47) that might provide complementary prognostic information. The broad implementation of such novel biomarkers in routine care is, however, hindered by the fact that research technologies are required for their measurements, such as mass spectrometry for the ceramide analysis and a modified aptamer-protein binding technology for the recently presented 9-protein based risk score. Moreover, the incremental prognostic importance of these and other new biomarkers needs to be evaluated and validated in large cohorts.

Patients with stable CHD carry not only a risk of CV death, but also a substantial risk of CV morbidity, which affects the lives of patients and their families and contributes to substantial health care costs (2). As expressed in the Salzburg statement, important health care decisions should be shared between the patient and caregiver (48). To make informed decisions, the risk of adverse events should be assessed and communicated with the patient. Therefore, it is an advantage that the present biomarker-based ABC-CHD model predicted not only the risk of CV death, but also the risk of most other important CV events, such as HF, MI, stroke, or composites of these events. Another advantage is that the biomarkers included in the ABC-CHD score are specific indicators of underlying CV pathophysiology, are readily available in routine care, and are not related to non-CV or cancer deaths (22). Considering the robust performance of

the novel ABC-CHD model, it can become a useful tool to guide decisions on more intense secondary prevention measures and to identify patients with unmet medical needs and elevated risk of adverse outcome for inclusion in future clinical trials of emerging treatments.

STUDY LIMITATIONS. Patients were included in a randomized controlled trial, which always involves a selection of patients compared with routine clinical practice; this is especially pertinent given that the trial employed risk-enrichment criteria. However, the model's robust performance in the external validation cohort is reassuring. Also, although the STABILITY trial was a global study, patients were well managed and the validation cohort included only patients in Germany, which could limit the generalizability to less developed health care systems. Additional validation in other cohorts could be of value. Finally, we did not have echocardiographic measures of LV function in this study. Thus, we cannot exclude that some of the patients had asymptomatic LV dysfunction.

CONCLUSIONS

In patients with stable CHD, the cardiac biomarkers NT-proBNP and hs-cTnT had greater prognostic value than any clinical variable or any other currently used biomarker for CV outcomes; plus, they provided incremental predictive information in addition to clinical variables and other biomarkers. Based on these findings, we developed and internally and externally validated a novel biomarker-based model for the prediction of CV death in patients with stable CHD containing age (A); the biomarkers (B) NT-proBNP, hs-cTnT, and LDL-C; and the clinical variables (C) smoking, DM, and prior PAD. The ABC-CHD score showed excellent discriminatory ability for CV death and other CV outcomes and was validated and well calibrated for CV death in an external cohort. Therefore, this novel biomarker-based ABC-CHD risk score might serve as a clinically useful decision support tool in patients with stable CHD.

ACKNOWLEDGMENTS The authors thank Vendela Roos and Ebba Bergman at Uppsala Clinical Research center for editorial support.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: A model combining patient age, clinical variables (smoking, DM, and PAD), and levels of biomarkers (NT-proBNP, hs-cTnT, and LDL-C) can be used to predict adverse CV outcomes including death.

TRANSLATIONAL OUTLOOK: Prospective studies are needed to establish the utility of this model for decision support in secondary prevention of CHD.

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KEY WORDS cardiac troponin
low-density lipoprotein cholesterol,
N-terminal pro-B-type natriuretic peptide,
risk prediction

APPENDIX For the expanded biomarker methods and TRIPOD checklist, as well as supplemental tables and figures, please see the online version of this paper.